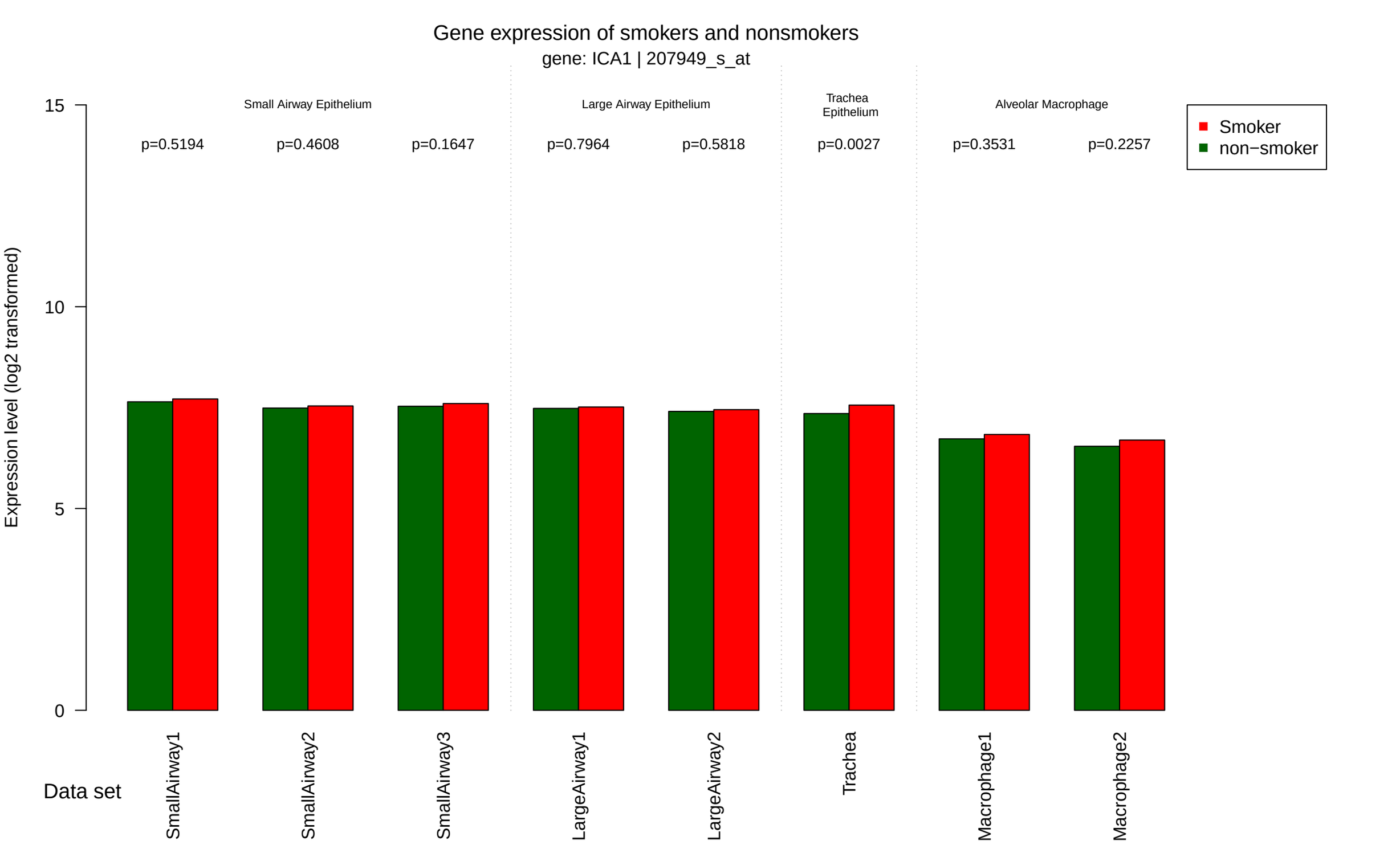
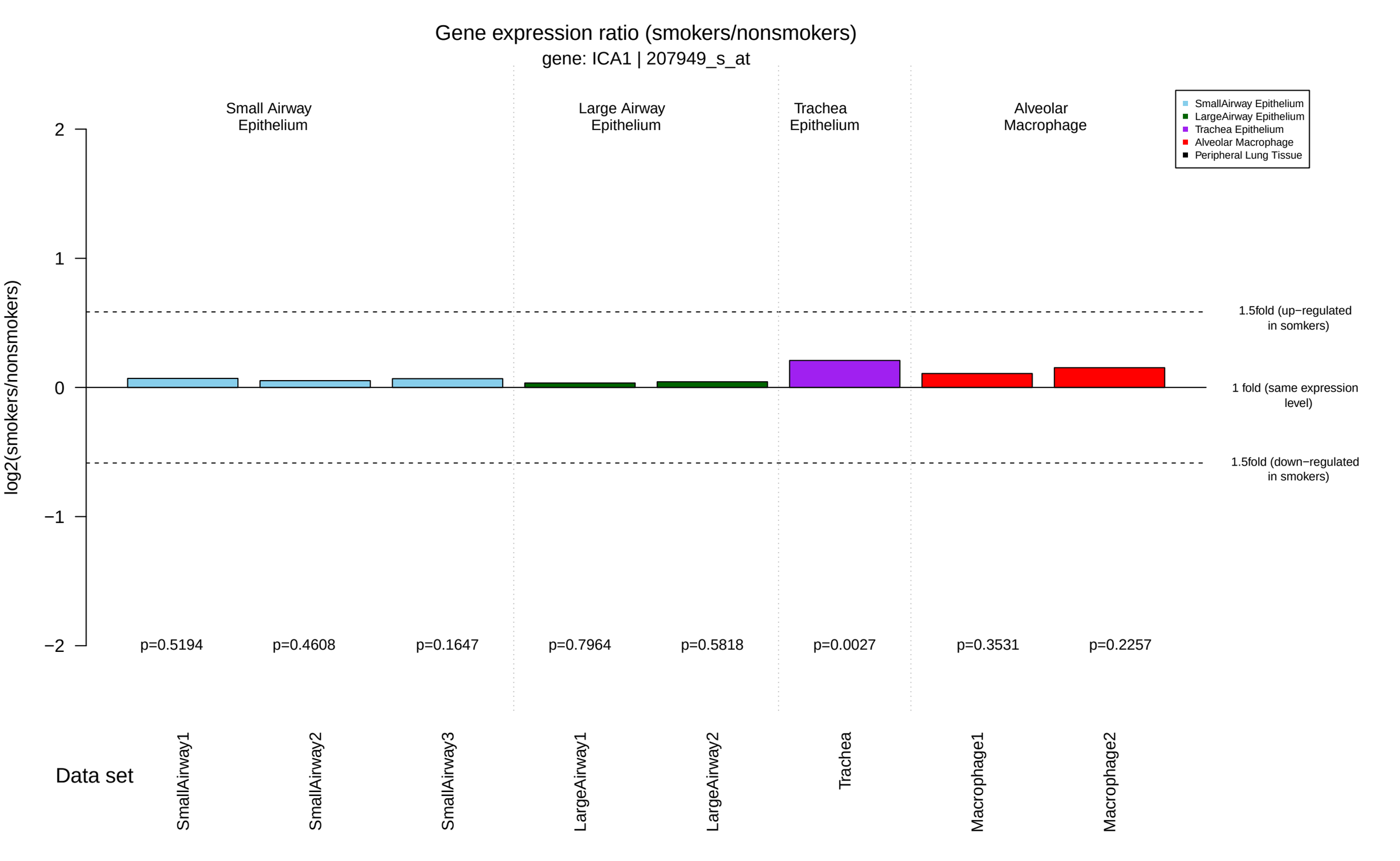
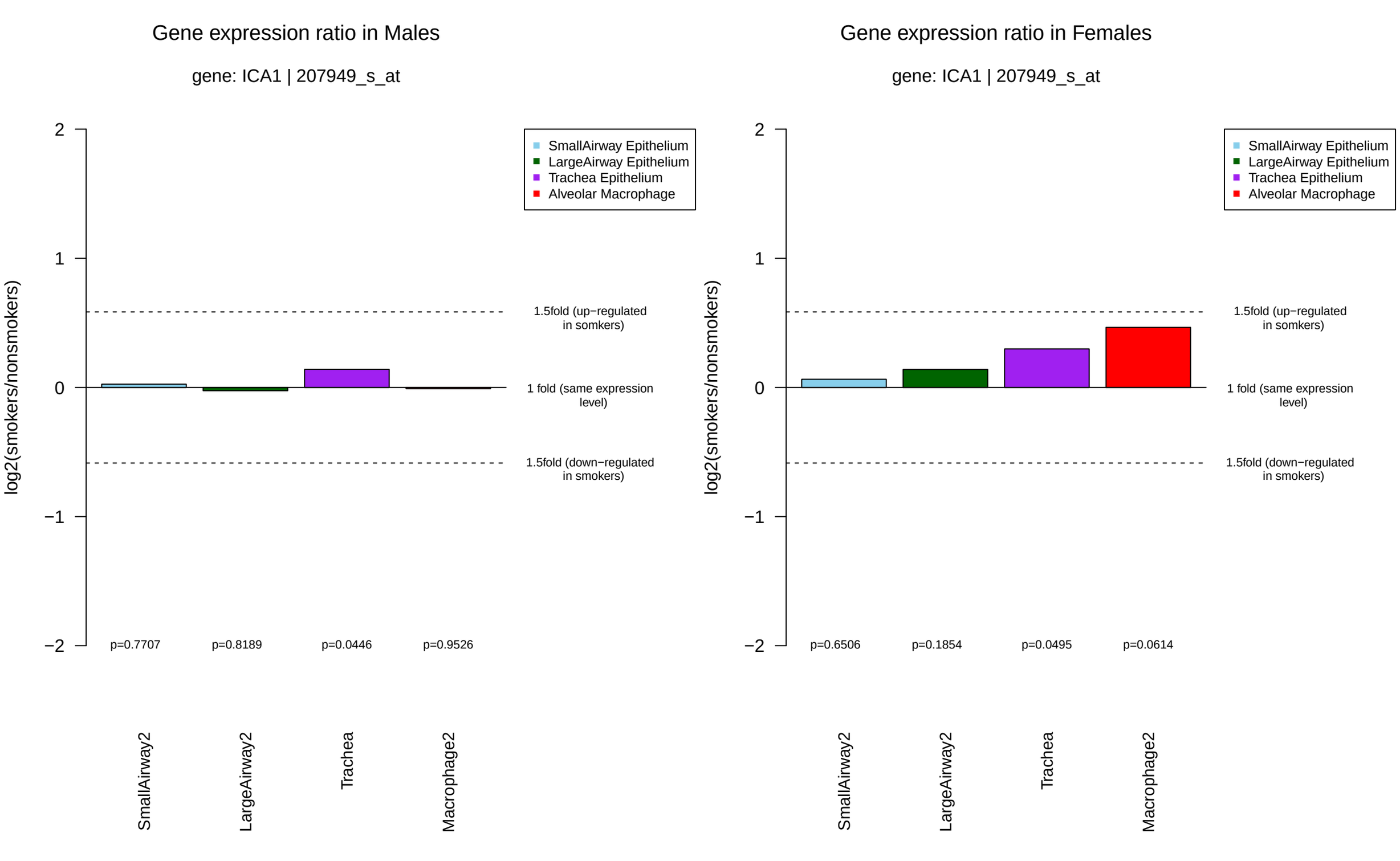
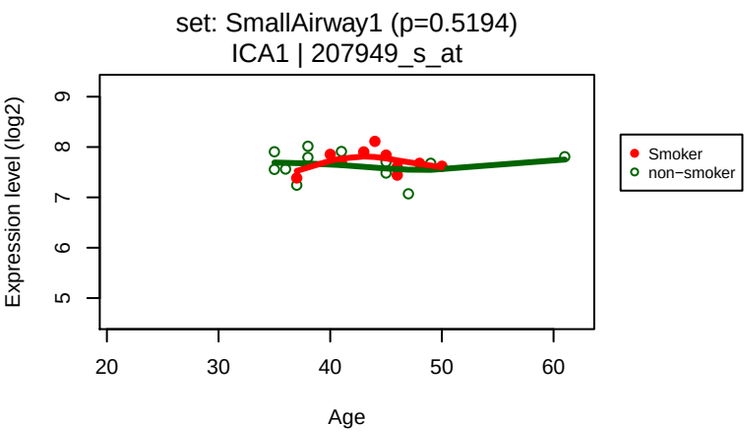
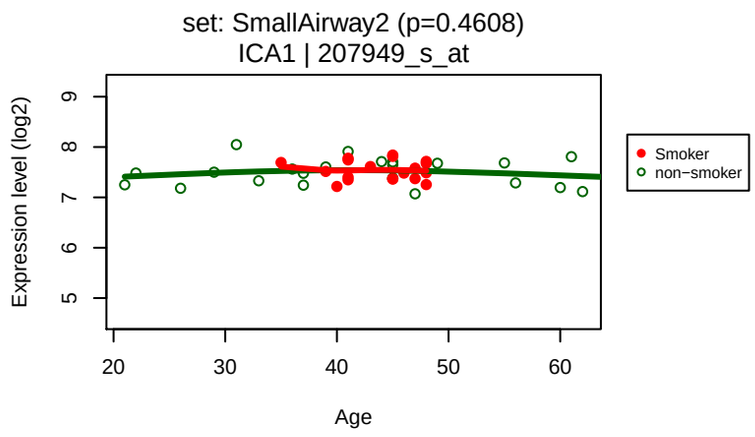
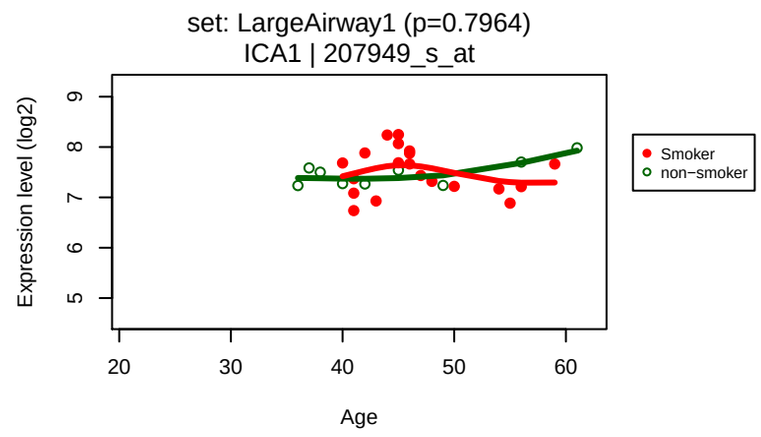

Gene expression of smokers and nonsmokers
gene: ICA1 | 207949_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5194
p=0.4608
p=0.1647
p=0.7964
p=0.5818
p=0.0027
p=0.3531
p=0.2257
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

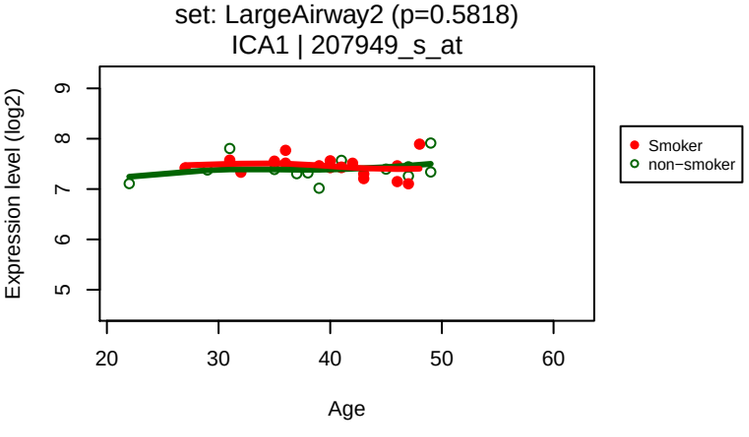

Gene expression ratio (smokers/nonsmokers)
gene: ICA1 | 207949_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5194
p=0.4608
p=0.1647
p=0.7964
p=0.5818
p=0.0027
p=0.3531
p=0.2257
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

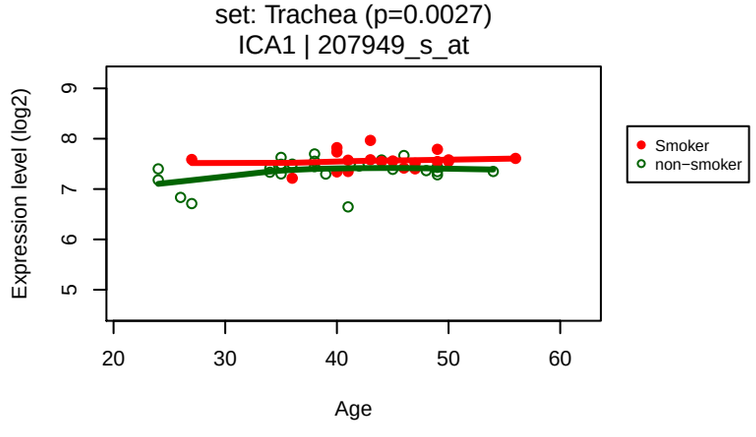

Gene expression ratio in Males
gene: ICA1 | 207949_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7707
p=0.8189
p=0.0446
p=0.9526
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ICA1 | 207949_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6506
p=0.1854
p=0.0495
p=0.0614
SmallAirway2
LargeAirway2
Trachea
Macrophage2

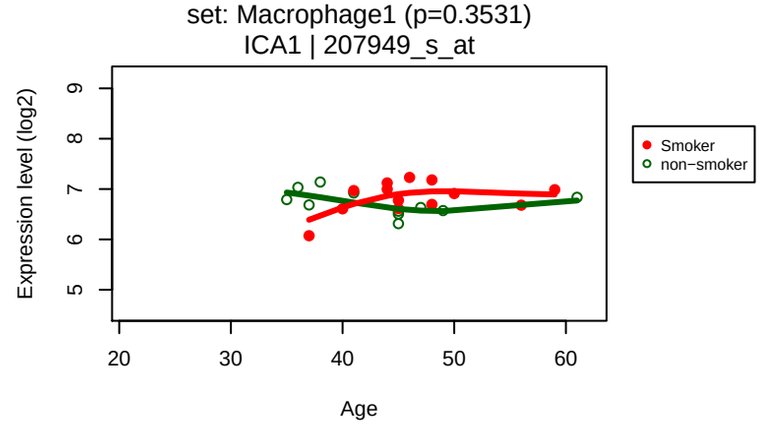

set: SmallAirway1 (p=0.5194)
ICA1 | 207949_s_at
Expression level (log2)
Age
Smoker
non−smoker

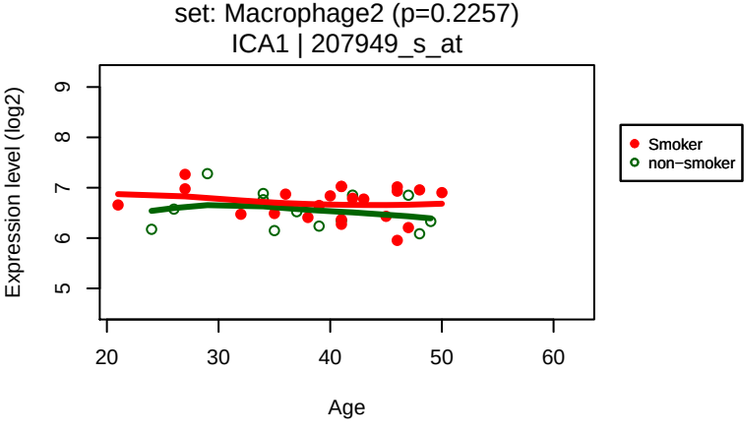

set: SmallAirway2 (p=0.4608)
ICA1 | 207949_s_at
Expression level (log2)
Age
Smoker
non−smoker

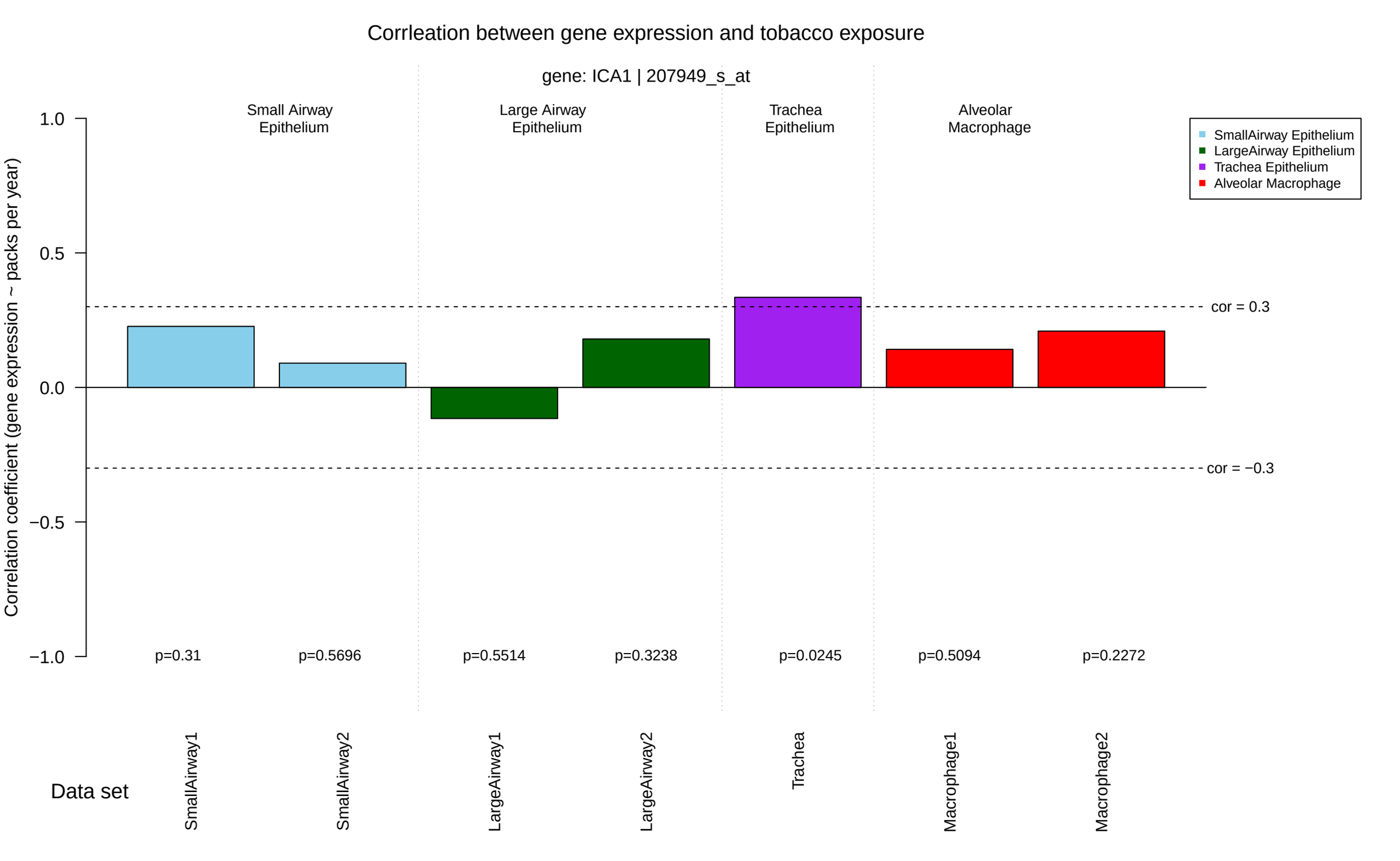

set: LargeAirway1 (p=0.7964)
ICA1 | 207949_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5818)
ICA1 | 207949_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0027)
ICA1 | 207949_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3531)
ICA1 | 207949_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2257)
ICA1 | 207949_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ICA1 | 207949_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.31
p=0.5696
p=0.5514
p=0.3238
p=0.0245
p=0.5094
p=0.2272
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2